

HDOC-9

Question-1:

Write a menu-driven C program to input two numbers and swap them using a third variable & without using third variable. Also display the values before and after swapping.

Task1: Algorithm:

1. Start.
2. Declare a, b, temp, and choice.
3. Input two numbers a and b.
4. Display the menu:
 - 1. Swap using third variable
 - 2. Swap without third variable
5. Input the user's choice.
6. Display a and b before swapping.
7. If choice is 1:
 - Store a in temp.
 - Assign b to a.
 - Assign temp to b.
8. Else if choice is 2:
 - $a = a + b$
 - $b = a - b$
 - $a = a - b$
9. Otherwise, display "Invalid choice".
10. Display a and b after swapping.
11. Stop.

Task2: pseudocode:

START

INPUT a, b

DISPLAY "1. Swap using third variable"

DISPLAY "2. Swap without third variable"

INPUT choice

DISPLAY a and b before swapping

IF choice = 1 THEN

temp = a

a = b

b = temp

ELSE IF choice = 2 THEN

a = a + b

b = a - b

a = a - b

ELSE

DISPLAY "Invalid choice"

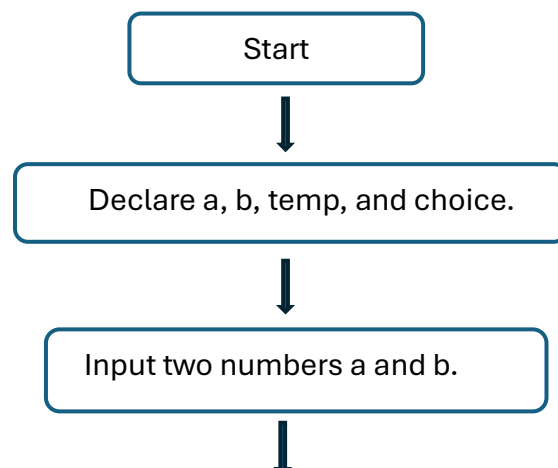
STOP

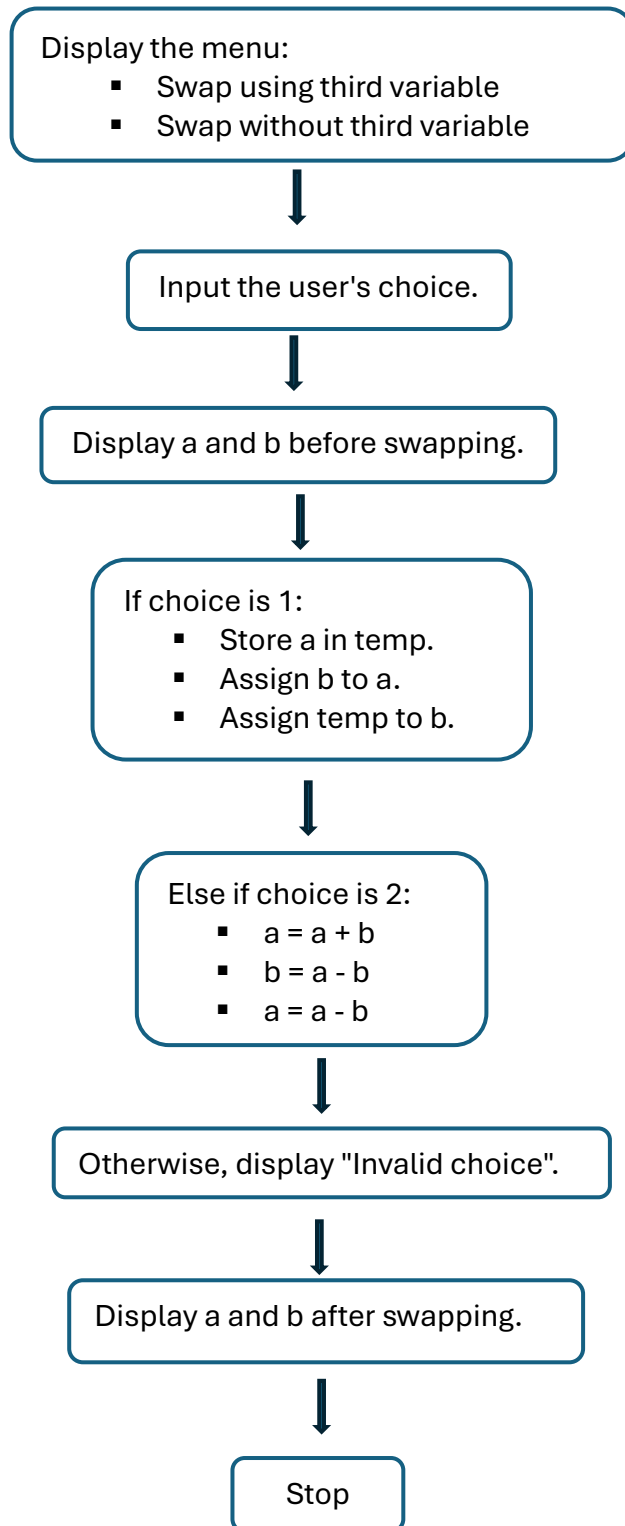
END IF

DISPLAY a and b after swapping

STOP

Task3: Flowchart:





Task4: Test case table:

Test Case	Input	Choice	Expected Output	Actual Output	Result
1	a=6, b=7	2	a=7, b=6	a=7, b=6	PASS
2	a=53, b=21	1	a=21, b=53	A=21, b=53	PASS
3	a=4, b=16	2	a=16, b=4	a=16, b=4	PASS
4	a=-3, b=7	1	a=7, b=-3	a=7, b=-3	PASS

Task5: C program:

```
#include <stdio.h>
int main()
{
    int a, b, temp, choice;
    printf("Enter two numbers: ");
    scanf("%d %d", &a, &b);
    printf("\nBefore Swapping: ");
    printf("\na = %d, b = %d\n", a, b);
    printf("\nMENU");
    printf("\n1. Swap using a third variable");
    printf("\n2. Swap without using third variable");
    printf("\nEnter your choice: ");
    scanf("%d", &choice);
    switch(choice)
    {
        case 1:
            temp = a;
            a = b;
            b = temp;
            printf("\nAfter Swapping: ");
            printf("\na = %d, b = %d\n", a, b);
            break;
        case 2:
            a = a + b;
            b = a - b;
            a = a - b;
            printf("\nAfter Swapping: ");
            printf("\na = %d, b = %d\n", a, b);
            break;
        default:
            printf("\nInvalid choice!\n");
    }
    return 0;
}
```

}

Task6: Output:

```
(kali㉿kali)-[~/HDOC9.c]
$ ./a.out
Enter two numbers: 6
7

Before Swapping:
a = 6, b = 7

MENU
1. Swap using a third variable
2. Swap without using third variable
Enter your choice: 2

After swapping:
a = 7, b = 6
```

```
(kali㉿kali)-[~/HDOC9.c]
$ ./a.out
Enter two numbers: 53
21

Before Swapping:
a = 53, b = 21

MENU
1. Swap using a third variable
2. Swap without using third variable
Enter your choice: 1

After Swapping:
= 21, b = 53
```

```
(kali㉿kali)-[~/HDOC9.c]
$ ./a.out
Enter two numbers: 4
16

Before Swapping:
a = 4, b = 16

MENU
1. Swap using a third variable
2. Swap without using third variable
Enter your choice: 2

After swapping:
a = 16, b = 4
```

```
(kali@kali)-[~/HDOC9.c]
$ ./a.out
Enter two numbers: -3
7

Before Swapping:
a = -3, b = 7

MENU
1. Swap using a third variable
2. Swap without using third variable
Enter your choice: 1

After Swapping:
a = 7, b = -3
```

Question 2:

Write a **menu-driven** C program to find the sum of the first n natural numbers and sum of squares of the first n natural numbers.

Task1: Algorithm:

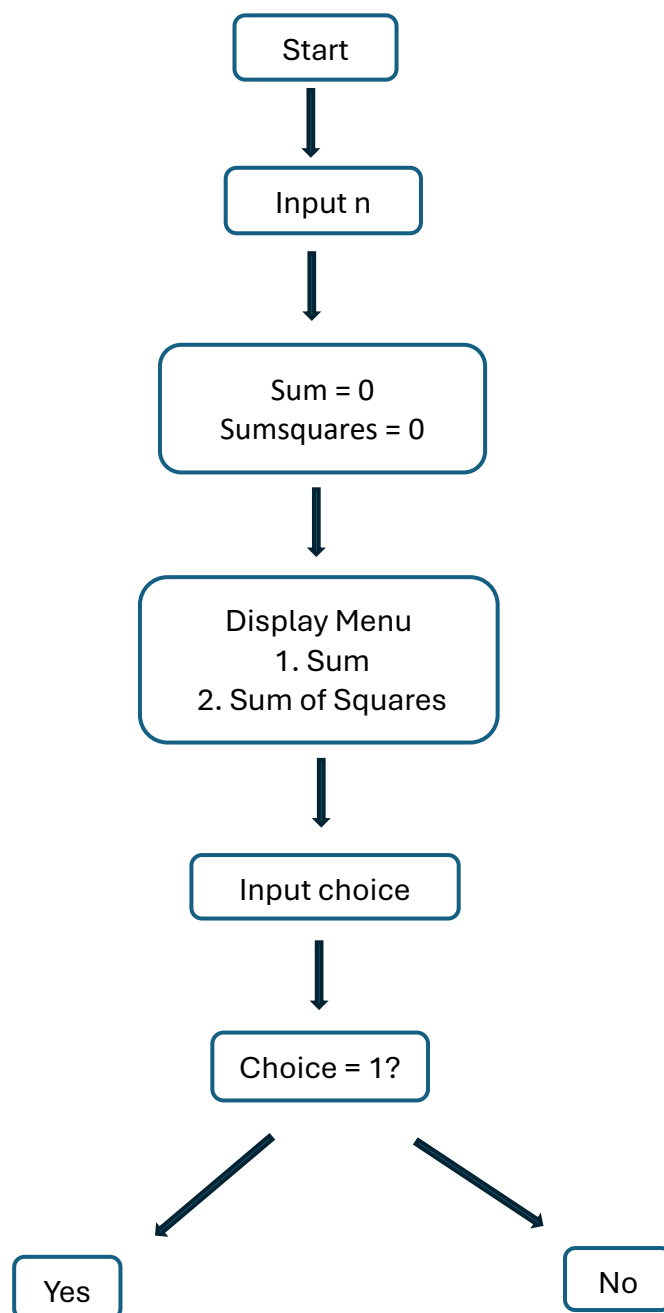
1. Start.
2. Input n.
3. Initialize sum = 0 and sumSquares = 0.
4. Display the menu.
5. Input the user's choice.
6. If choice is 1:
 - o Repeat from i = 1 to n.
 - o Calculate sum = sum + i.
 - o Display sum.
7. Else if choice is 2:
 - o Repeat from i = 1 to n.
 - o Calculate sumSquares = sumSquares + i × i.
 - o Display sum of squares.
8. Otherwise, display "Invalid choice".
9. Stop.

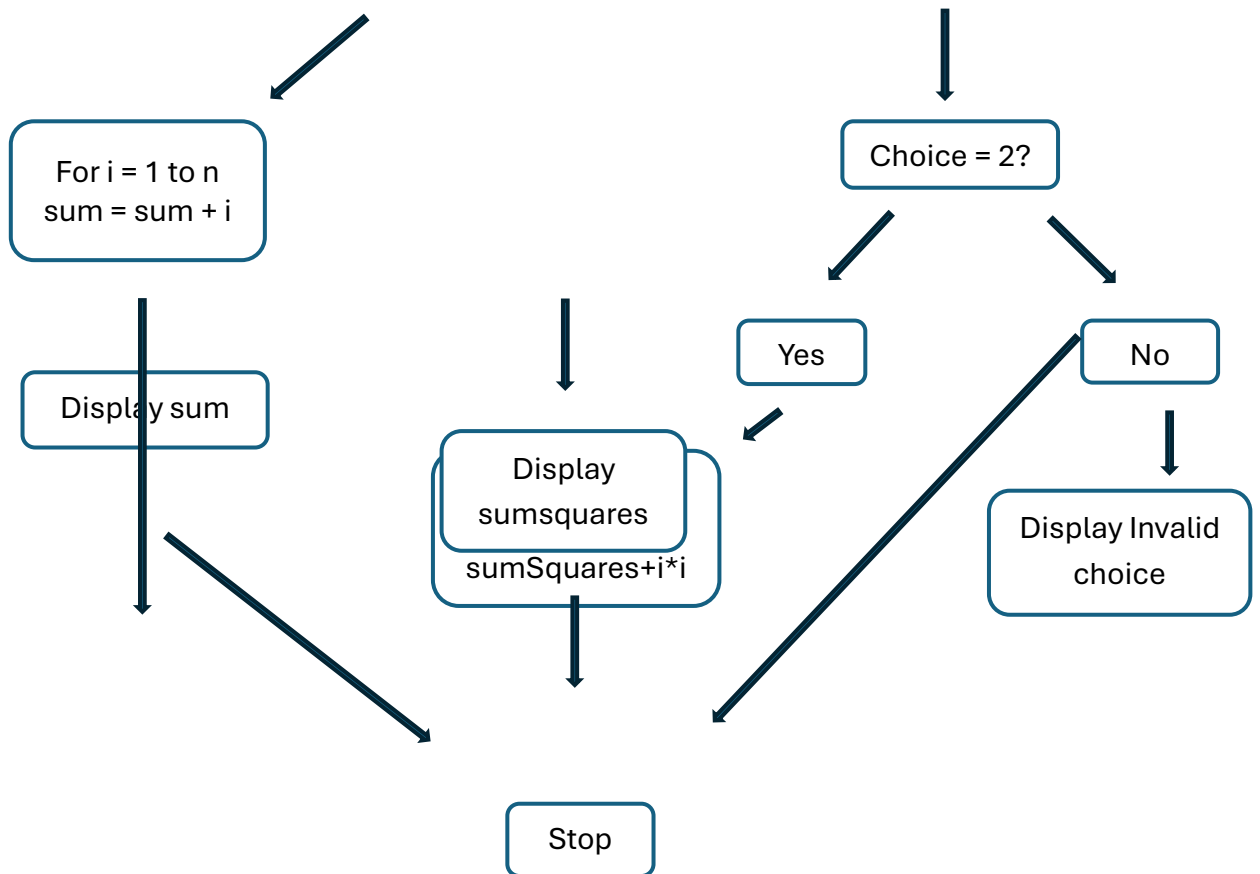
Task2: pseudocode:

- START
- INPUT n
- SET sum = 0
- SET sumSquares = 0
- DISPLAY "MENU"
- DISPLAY "1. Sum of natural numbers"
- DISPLAY "2. Sum of squares"
- INPUT choice
- IF choice = 1 THEN
 - FOR i = 1 TO n
 - sum = sum + i

- END FOR
- DISPLAY sum
- ELSE IF choice = 2 THEN
 FOR i = 1 TO n
 SumSquares = sumSquares + (i * i)
- END FOR
- DISPLAY sumSquares
- ELSE
- DISPLAY "Invalid Choice"
- END IF
- STOP

Task3: Flowchart:





Task4: Test case table:

Test Case	Input n	Choice	Expected Output	Actual Output	Result
1	3	1	Sum = 6	Sum = 6	PASS
2	14	2	Sum of squares = 1015	Sum of squares = 1015.	PASS
3	10	1	Sum = 55	Sum = 55	PASS
4	6	2	Sum of squares = 91	Sum of squares = 91	PASS

Task5: C program:

```
#include <stdio.h>
int main()
{
    int n, choice, i;
    int sum = 0, sumSquares = 0;
    printf("Enter the value of n: ");
    scanf("%d", &n);
    printf("\nMENU");
    printf("\n1. Sum of first n natural numbers");
    printf("\n2. Sum of squares of first n natural numbers");
    printf("\nEnter your choice: ");
    scanf("%d", &choice);
    switch(choice)
    {
        case 1:
            for(i = 1; i <= n; i++)
            {
                sum = sum + i;
            }
            printf("\nSum of first %d natural numbers = %d\n", n, sum);
            break;
        case 2:
            for(i = 1; i <= n; i++)
            {
                sumSquares = sumSquares + (i * i);
            }
            printf("\nSum of squares of first %d natural numbers = %d\n", n, sumSquares);
            break;
        default:
            printf("\nInvalid choice!\n");
    }
    return 0;
}
```

Task6: Output:

```
(kali㉿kali)-[~/HDOC9.c]
$ ./a.out
Enter the value of n: 3

MENU
1. Sum of first n natural numbers
2. Sum of squares of first n natural nummbers
Enter your choice: 1

Sum of first 3 natural numbers = 6
```

```
(kali㉿kali)-[~/HDOC9.c]
$ ./a.out
Enter the value of n: 14

MENU
1. Sum of first n natural numbers
2. Sum of squares of first n natural nummbers
Enter your choice: 2

Sum of squares of first 14 natural numbers = 1015
```

```
(kali㉿kali)-[~/HDOC9.c]
$ ./a.out
Enter the value of n: 10

MENU
1. Sum of first n natural numbers
2. Sum of squares of first n natural nummbers
Enter your choice: 1

Sum of first 10 natural numbers = 55
```

```
(kali㉿kali)-[~/HDOC9.c]
$ ./a.out
Enter the value of n: 6

MENU
1. Sum of first n natural numbers
2. Sum of squares of first n natural nummbers
Enter your choice: 2

Sum of squares of first 6 natural numbers = 91
```

Question 3:

Write a **menu-driven** C program to calculate simple interest, compound amount, compound interest, and total payable amount for given principal, rate, and time.

Task1: Algorithm:

1. Start.
2. Declare P, R, T, SI, A, CI, total, and choice.
3. Input the principal amount P.
4. Input the rate of interest R.
5. Input the time T.
6. Display the menu with four options.
7. Input choice.
8. If choice is **1**, calculate simple interest and display it.
9. If choice is **2**, calculate compound amount and display it.
10. If choice is **3**, calculate compound amount, then calculate compound interest and display it.
11. If choice is **4**, calculate simple interest, add it to the principal, and display the total payable amount.
12. For any other choice, display "Invalid choice".
13. Stop.

Task2: pseudocode:

Start

INPUT P, R, T

DISPLAY "MENU"

DISPLAY "1. Simple Interest"

DISPLAY "2. Compound Amount"

DISPLAY "3. Compound Interest"

DISPLAY "4. Total Payable Amount"

INPUT choice

SWITCH choice

CASE 1:

$$SI = (P * R * T) / 100$$

DISPLAY SI

CASE 2:

$$A = P * (1 + R/100)^T$$

DISPLAY A

CASE 3:

$$A = P * (1 + R/100)^T$$

$$CI = A - P$$

DISPLAY CI

CASE 4:

$$SI = (P * R * T) / 100$$

$$\text{total} = P + SI$$

DISPLAY total

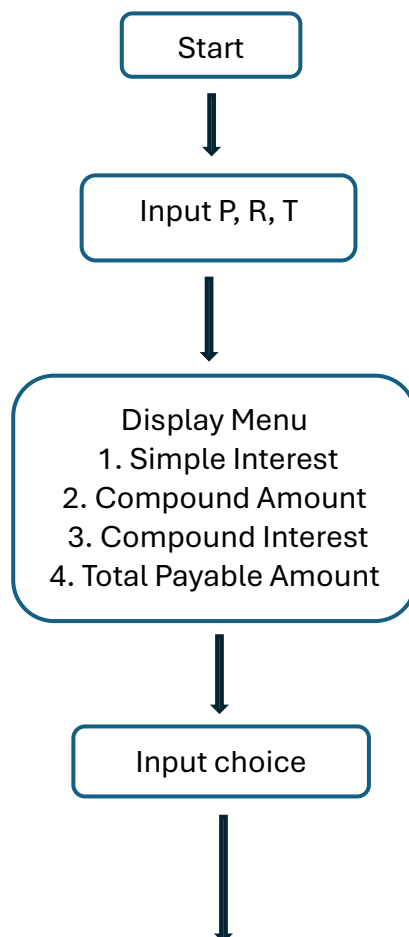
DEFAULT:

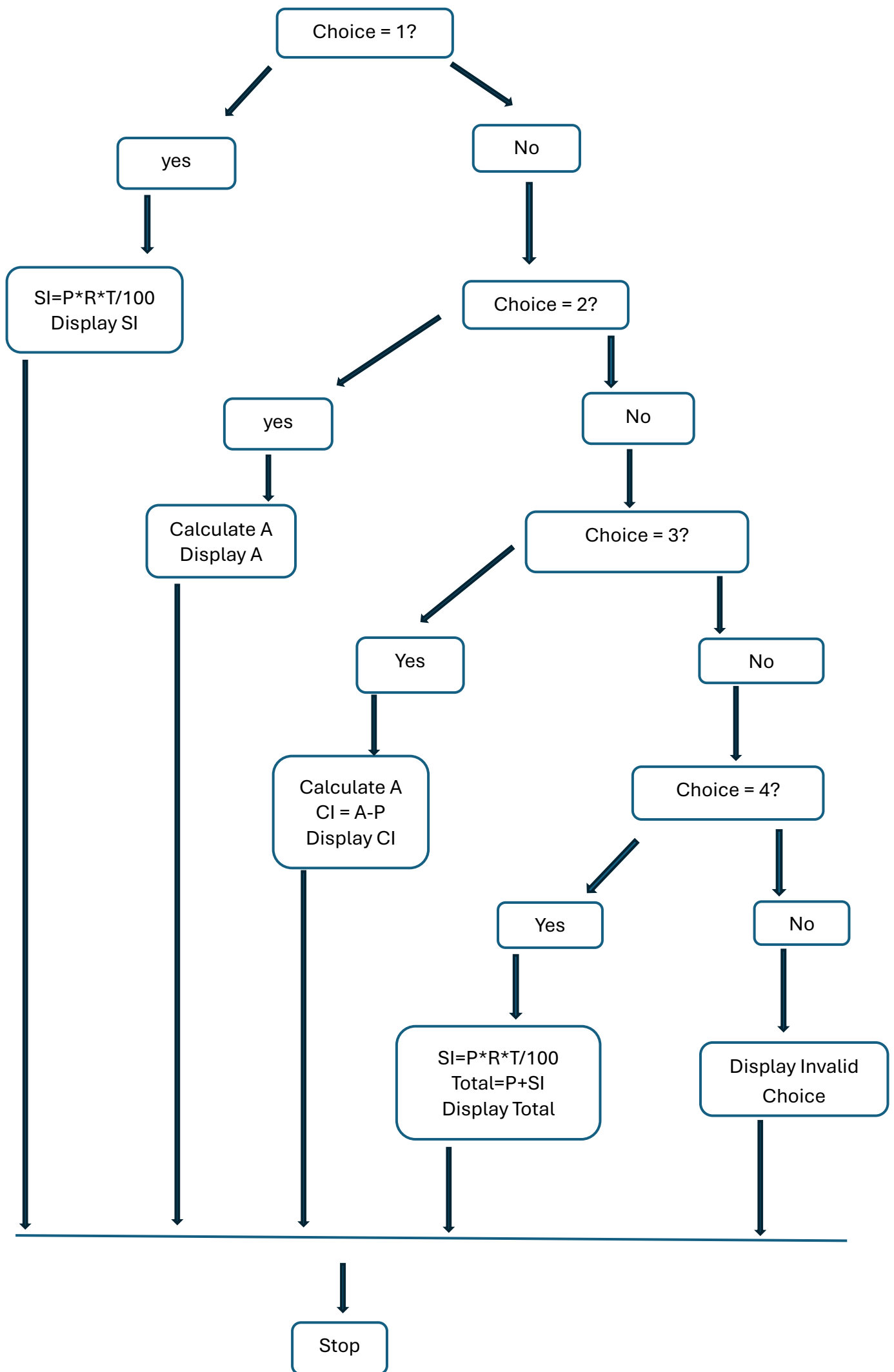
DISPLAY "Invalid Choice"

End switch

STOP

Task3: Flowchart:





Task4: Test case table:

Test Case	P	R	T	Choice	Expected Output	Actual Output	Result
1	1000	10	2	1	SI = 200.00	SI = 200.00	PASS
2	1000	10	2	2	Amount = 1210.00	Amount = 1210.00	PASS
3	1000	10	2	3	CI = 210.00	CI = 210.00	PASS
4	1000	10	2	4	Total = 1200.00	Total = 1200.00	PASS

Task5: C program:

```
#include <stdio.h>
#include <math.h>
int main()
{
    float P, R, T;
    float SI, A, CI, total;
    int choice;

    printf("Enter Principal Amount: ");
    scanf("%f", &P);
    printf("Enter Rate of Interest: ");
    scanf("%f", &R);
    printf("Enter Time in years: ");
    scanf("%f", &T);
    printf("\nMENU");
    printf("\n1. Simple Interest");
    printf("\n2. Compound Amount");
    printf("\n3. Compound Interest");
    printf("\n4. Total Payable Amount");
    printf("\nEnter your choice: ");
    scanf("%d", &choice);

    switch(choice)
```

```

{
    case 1:
        SI = (P * R * T) / 100;
        printf("\nSimple Interest = %.2f\n", SI);
        break;
    case 2:
        A = P * pow((1 + R / 100), T);
        printf("\nCompound Amount = %.2f\n", A);
        break;
    case 3:
        A = P * pow((1 + R / 100), T);
        CI = A - P;
        printf("\nCompound Interest = %.2f\n", CI);
        break;
    case 4:
        SI = (P * R * T) / 100;
        total = P + SI;
        printf("\nTotal Payable Amount = %.2f\n", total);
        break;
    default:
        printf("\nInvalid choice\n");
}
return 0;
}

```

Task6: Output:

```

(kali㉿kali)-[~/HDOC9.c]
$ ./a.out
Enter principle amount: 1000
Enter rate of intrest: 10
Enter time in years: 2

MENU
1. Simple interest
2. Compound amount
3. Compound Interest
4. Total payable amount
Enter your choice: 1

Simple Interest = 200

```

```
(kali㉿kali)-[~/HDOC9.c]
$ ./a.out
Enter principle amount: 1000
Enter rate of intrest: 10
Enter time in years: 2
```

```
MENU
1. Simple interest
2. Compound amount
3. Compound Interest
4. Total payable amount
Enter your choice: 2
```

```
Compound amount = 1210
```

```
(kali㉿kali)-[~/HDOC9.c]
$ ./a.out
Enter principle amount: 1000
Enter rate of intrest: 10
Enter time in years: 2
```

```
MENU
1. Simple interest
2. Compound amount
3. Compound Interest
4. Total payable amount
Enter your choice: 3
```

```
Compund interest = 210
```

```
(kali㉿kali)-[~/HDOC9.c]
$ ./a.out
Enter principle amount: 1000
Enter rate of intrest: 10
Enter time in years: 2
```

```
MENU
1. Simple interest
2. Compound amount
3. Compound Interest
4. Total payable amount
Enter your choice: 4
```

```
Total Payable Amount = 1200
```